



Physics I-SEE-R Problem-Solving Worksheet

Identify → Simplify → Equation → Execute → Reflect

This worksheet is a mechanical procedure. It does the thinking so you do not have to. Fill every blank in order. Do not skip steps. The protocol catches your errors before you box the answer.

Reference: `honors-physics-expression-to-equation-vocabulary.md` (430 cards)

Method: `physics-without-intuition.md`

Problem ID	Difficulty Level	Domain(s)	Time Spent
	Level 1 / 2 / 3 / 4		_____ min

Level guide: Level 1 = 1–3 cards (direct substitution). Level 2 = 2–5 cards (one bridge variable). Level 3 = 3–8 cards (multi-context or variable grouping required). Level 4 = 5–15 cards (3+ domain synthesis).

PRE-METHOD — The Four Contracts

"Establish these before touching a single equation. Never change them mid-problem."

Physics Without Intuition, §0.1

Coordinate System

Axis	Direction	Why This Direction?
$+x$		Direction of primary motion
$+y$		Upward, or perpendicular away from incline surface
Origin		Object's starting position, or problem's reference point
$t = 0$ at		Instant action begins: release / throw / collision

Golden Rule: Once written, never change. A negative answer means "opposite the + direction." It does not mean "wrong."

STEP I — IDENTIFY

"Which cards fire? What is given? What is hidden?"

Physics Without Intuition, §0.4 — The 4-Pass Extraction

Pass 1 — Five-Word Gist (pencil down)

Read once. No writing. State the core of the problem in five words or fewer.

Pass 2 — Variable Table

Extract every number with its unit. Implied zeros are not optional — they get their own rows. If you miss one, you have too many unknowns and cannot solve. Count knowns vs. unknowns after

completing the table.

#	Variable Name	Symbol	Value	Unit	Dimension	Source
1						
2						
3						
4						
5						
6						
7						
8						
TARGET			?			

Knowns: ____ | **Unknowns:** ____ | (Must have: unknowns $\leq$ knowns + equations from cards)

Implied Zero Scan

Protocol A from Physics Without Intuition, Part VIII. Check every box below BEFORE moving to Pass 3.

If the problem says...	Then...	Card	Found?	Added to table?
"from rest" / "dropped" / "released"	$v_0 = 0$	1, 308	☐	☐
"stops" / "comes to rest" / "brakes"	$v = 0$	2, 309	☐	☐
"maximum height" / "peak" / "top" / "apex"	$v_y = 0$	13, 310	☐	☐

If the problem says...	Then...	Card	Found?	Added to table?
"frictionless" / "smooth" / "ice"	$f = 0$	31, 311	[]	[]
"constant velocity" / "steady speed" / "coasting"	$a = 0$	3, 318	[]	[]
"free fall" / "falling freely"	$a_y = -g$	6	[]	[]
"just loses contact" / "leaves surface"	$N = 0$	93, 316	[]	[]
"massless rope" / "ideal string"	T uniform	34, 312	[]	[]
"equilibrium" / "at rest" (with forces)	$\sum F = 0$	25, 319	[]	[]
"level ground" (projectile)	$\Delta y = 0$	18, 314	[]	[]
"terminal velocity"	$\sum F = 0$	47, 320	[]	[]
"elastic collision"	$K_i = K_f$	74	[]	[]
"stick together" / "embedded"	One shared v_f	73	[]	[]
"rolling without slipping"	$v = \omega R$	113	[]	[]

Pass 3 — Trigger Trace

Underline every physics trigger phrase in the problem statement. For each one, write the card number and the instant equation that card demands. The cards choose — you transcribe. If the problem crosses domains, the order of cards tells you the solution architecture.

Order	Trigger Phrase in Problem	Card #	Instant Equation
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			

Cards fired: ____ | Domain chain: _____

Difficulty: 1–3 cards = Level 1. 4–5 = Level 2. 6–8 = Level 3. 9+ = Level 4.

Pass 4 — Unknown

What am I solving for?	Symbol	Unit	Target Dimension

Bridge Recognition

Physics Without Intuition, Part V. Two or more distinct events separated in time?

Does this problem chain multiple domains? ☐ Yes ☐ No

If YES — what does the first event produce that the second event needs?

First event involves...	Bridge domain	Output	Bridge card
[] Collision / explosion / recoil	Momentum	v_f	321–327
[] Charged particle + voltage / E-field	Energy ($\frac{1}{2}mv^2 = qV$)	v	338–344
[] Falling / sliding / spring	Energy conservation	v or x	328–337
[] Gas changing P, V, T	Thermo	P, V, T, W	350–354
[] Acceleration over distance/time	Kinematics	v or t	—

Bridge variable: _____ | Second domain: _____

STEP S — SIMPLIFY

"Cross out what does not matter. Every assumption you can justify removes one unknown."

Assumption	Condition	If true, then...	[]
No friction	"frictionless", "smooth", "ice"	$f = 0$, mechanical energy conserved	[]
Massless rope/pulley	"ideal", "light", "massless"	T uniform	[]
No air resistance	Not mentioned, or "neglect"	Only gravity acts	[]
1D motion	Only one axis changes	Ignore perpendicular axis	[]

Assumption	Condition	If true, then...	[]
Point mass	No rotation mentioned	No moment of inertia	[]
Ideal gas	No "real gas" / "van der Waals"	$PV = nRT$ applies	[]
Light spring	Spring mass not given	Spring mass = 0	[]
Inextensible string	Not "elastic"	a same for connected objects	[]
Small angle	θ small (pendulum)	$\sin \theta \approx \theta$	[]
Isolated system	"no external forces"	Momentum conserved	[]
Conservative only	No friction, no drag	E_{mech} conserved	[]

Additional simplifications:

STEP E — EQUATION

"The cards chose. Now write them in order. If the problem crosses domains, the first equation's output feeds the second."

Step	Card #	Equation	Purpose
1			
2			
3			
4			
5			
6			
7			
8			

Primary governing principle: _____

If kinematics: Which variable am I NOT given and do NOT need?

Missing variable	→ Equation

STEP E — EXECUTE

"Keep symbols until the target is isolated. Substitute numbers last. This catches algebra errors early and reveals cancellations."

Algebraic Solution (symbols only)

(Derive the expression for the target variable — no numbers yet)

Numerical Substitution

Step	Card #	Calculation	Result
1			
2			
3			
4			
5			
6			
7			
8			

Unit Trace

Follow the units from the first given quantity through every operation. If they do not cancel to the target unit, the equation path is wrong. Fix it before boxing the answer.

_____ → _____ → _____ → _____ → _____ → **TARGET**

Units cancel correctly? ☐ Yes ☐ No → Back to Equation step

Answer

Value: _____ **Unit:** _____ **Direction (if vector):** _____

Boxed answer: _____

STEP R — REFLECT

"Three mechanical checks. Pass all three before moving to the next problem."

Check 1: Dimensional Analysis (Card 430)

LHS (your answer's dimension)	RHS (target dimension)	Match?
		☐ Yes ☐ No

Check 2: Extreme Case Test

Test (set one variable to limit)	Expected behavior	Your formula predicts...	Pass?
			☐
			☐

Check 3: Sign Sanity

Quantity	Your sign	Expected (physical reasoning)	OK?
			☐

Error Log (complete only if answer was wrong)

"The error log is not a record of failure. It is a vaccine against repeating the same mistake."

Error type(s):

Category	Specific detail (which variable? which step?)	[]
Missed implied zero		[]
Coordinate sign error		[]
Wrong kinematic equation (missing-variable rule)		[]
FBD error (missing/wrong force direction)		[]
Trig decomposition error (sin vs cos)		[]
Conservation law misapplied		[]
Collision type misidentified		[]
Forgot non-conservative work term		[]
Unit mismatch (cm↔m, g↔kg, h↔s)		[]
Algebra error (sign, factor of 2)		[]
g sign error		[]
Tension direction wrong		[]
Normal force incorrectly set to mg		[]

Category	Specific detail (which variable? which step?)	[]
Bridge variable not identified		[]
Variable-Group Method needed		[]
Other:		[]

Root cause (one sentence):

Prevention (actionable, specific):

Post-Solve Summary

Item	Value
Cards fired	
Domains crossed	
Domain list (in order)	
Pivot card(s)	
FBD needed?	☐ Yes ☐ No
Conservation law?	☐ Energy ☐ Momentum ☐ Both ☐ Neither
Coordinate	☐ Cartesian ☐ Tilted ☐ Radial

Item	Value
system	
Quadratic?	☐ Yes ☐ No
Architecture	☐ Single-domain ☐ Two-domain bridge ☐ Multi-context ☐ 3+ domain synthesis

QUICK-REFERENCE PANEL

Keep this panel available while solving. Contains the most commonly needed data from the 430-card vocabulary deck.

Implied Zero Catalog

Phrase	Value	Card
"from rest" / "dropped" / "released"	$v_0 = 0$	1, 308
"stops" / "comes to rest"	$v = 0$	2, 309
"maximum height" / "peak" / "top"	$v_y = 0$	13, 310
"frictionless" / "smooth" / "ice"	$f = 0$	31, 311
"constant velocity" / "steady speed"	$a = 0$	3, 318
"free fall"	$a_y = -g$	6
"just loses contact"	$N = 0$	93, 316
"massless rope"	T uniform	34, 312
"equilibrium"	$\sum F = 0, \sum \tau = 0$	25, 319

Phrase	Value	Card
"level ground" (projectile)	$\Delta y = 0$	18, 314
"terminal velocity"	$a = 0, \sum F = 0$	47, 320
"elastic collision"	$K_i = K_f$	74
"stick together" / "embedded"	shared v_f	73
"rolling without slipping"	$v = \omega R$	113

Kinematic Equation Selector (Card 4)

I need...	I'm missing...	Use this
v	t	$v^2 = v_0^2 + 2a\Delta x$
v	Δx	$v = v_0 + at$
Δx	v	$\Delta x = v_0 t + \frac{1}{2}at^2$
Δx	a	$\Delta x = \frac{1}{2}(v_0 + v)t$
t	v	$\Delta x = v_0 t + \frac{1}{2}at^2$
t	Δx	$v = v_0 + at$
a	t	$v^2 = v_0^2 + 2a\Delta x$

Dimensional Analysis (Cards 401–430)

Quantity	Symbol	Dimension
Speed / velocity	v	$[L][T]^{-1}$

Quantity	Symbol	Dimension
Acceleration	a	$[L][T]^{-2}$
Force	F	$[M][L][T]^{-2}$
Energy / work / heat	E, K, U, W, Q	$[M][L]^2[T]^{-2}$
Momentum / impulse	p, J	$[M][L][T]^{-1}$
Power	P	$[M][L]^2[T]^{-3}$
Pressure	P	$[M][L]^{-1}[T]^{-2}$
Torque	τ	$[M][L]^2[T]^{-2}$
Moment of inertia	I	$[M][L]^2$
Angular velocity	ω	$[T]^{-1}$
Angular momentum	L	$[M][L]^2[T]^{-1}$
Spring constant	k	$[M][T]^{-2}$
Frequency	f	$[T]^{-1}$
Electric charge	q	$[Q]$
Electric field	E	$[M][L][Q]^{-1}[T]^{-2}$
Voltage	V	$[M][L]^2[Q]^{-1}[T]^{-2}$
Magnetic field	B	$[M][Q]^{-1}[T]^{-1}$
Resistance	R	$[M][L]^2[Q]^{-2}[T]^{-1}$

Physical Constants

Constant	Symbol	Value
g (Earth surface)	g	9.8 m/s ² (AP: 10)

Constant	Symbol	Value
Universal gravitation	G	$6.67 \times 10^{-11} \text{ N m}^2/\text{kg}^2$
Speed of light	c	$3.00 \times 10^8 \text{ m/s}$
Elementary charge	e	$1.60 \times 10^{-19} \text{ C}$
Electron mass	m_e	$9.11 \times 10^{-31} \text{ kg}$
Proton mass	m_p	$1.67 \times 10^{-27} \text{ kg}$
Coulomb constant	k	$8.99 \times 10^9 \text{ N m}^2/\text{C}^2$
ϵ_0	ϵ_0	$8.85 \times 10^{-12} \text{ C}^2/(\text{N m}^2)$
μ_0	μ_0	$4\pi \times 10^{-7} \text{ T m/A}$
Planck's constant	h	$6.63 \times 10^{-34} \text{ J s}$
Boltzmann constant	k_B	$1.38 \times 10^{-23} \text{ J/K}$
Avogadro's number	N_A	$6.022 \times 10^{23} \text{ mol}^{-1}$
Gas constant (J)	R	$8.314 \text{ J}/(\text{mol K})$
Gas constant (L·atm)	R	$0.08206 \text{ L atm}/(\text{mol K})$
Atmospheric pressure	P_{atm}	$1.013 \times 10^5 \text{ Pa} = 760 \text{ mmHg}$

Problem-Type Fast Routing

Problem opens with...	Domain	Cards
"accelerates" / "thrown" / no forces	Kinematics	1-22
"force" / "push" / "pull" / "friction" / "incline"	Forces	23-49
"work" / "speed at bottom" / "spring" / no time	Energy	50-68
"collision" / "hit" / "bullet" / "recoil" / "explosion"	Momentum	69-82

Problem opens with...	Domain	Cards
"circular" / "orbit" / "centripetal" / "banked"	Circular & Gravitation	83–101
"torque" / "angular" / "rolls" / "spins"	Rotational	102–120
"oscillates" / "pendulum" / "period"	SHM	121–131
"wave" / "Doppler" / "standing wave" / "harmonic"	Waves & Sound	132–153
"charge" / "Coulomb" / "E-field" / "capacitor"	Electrostatics	154–170
"circuit" / "current" / "resistor" / "Ohm"	Circuits	171–192
"magnetic" / "B-field" / "flux" / "induction"	Magnetism	193–216
"heat" / "entropy" / "Carnot" / "isothermal"	Thermodynamics	217–246
"fluid" / "buoyancy" / "Bernoulli" / "flow"	Fluids	247–260
"lens" / "mirror" / "refraction" / "Snell"	Optics	261–282
"photon" / "photoelectric" / "relativity" / "nuclear"	Modern Physics	283–307

"The exam is won before you enter the room. This worksheet is the bridge between your 430 flashcard reflexes and the mechanical solution of any problem."

— *Physics Without Intuition*

First 20 problems: Fill every section. It will feel slow — you are building the protocol.

Problems 20–50: The cycle accelerates. You will notice yourself anticipating the next step.

Problems 50+: The worksheet becomes a mental checklist. You will catch your own errors before you make them.

Difficulty = number of cards fired. Level 1 fires 3 cards. Level 4 fires 10. The method is identical. Count the cards. Trace the triggers. Execute.